Lecture 3(b): Flexibility classification

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Agenda

- Flexibility classification
- What is fairness?
- Notions of fairness

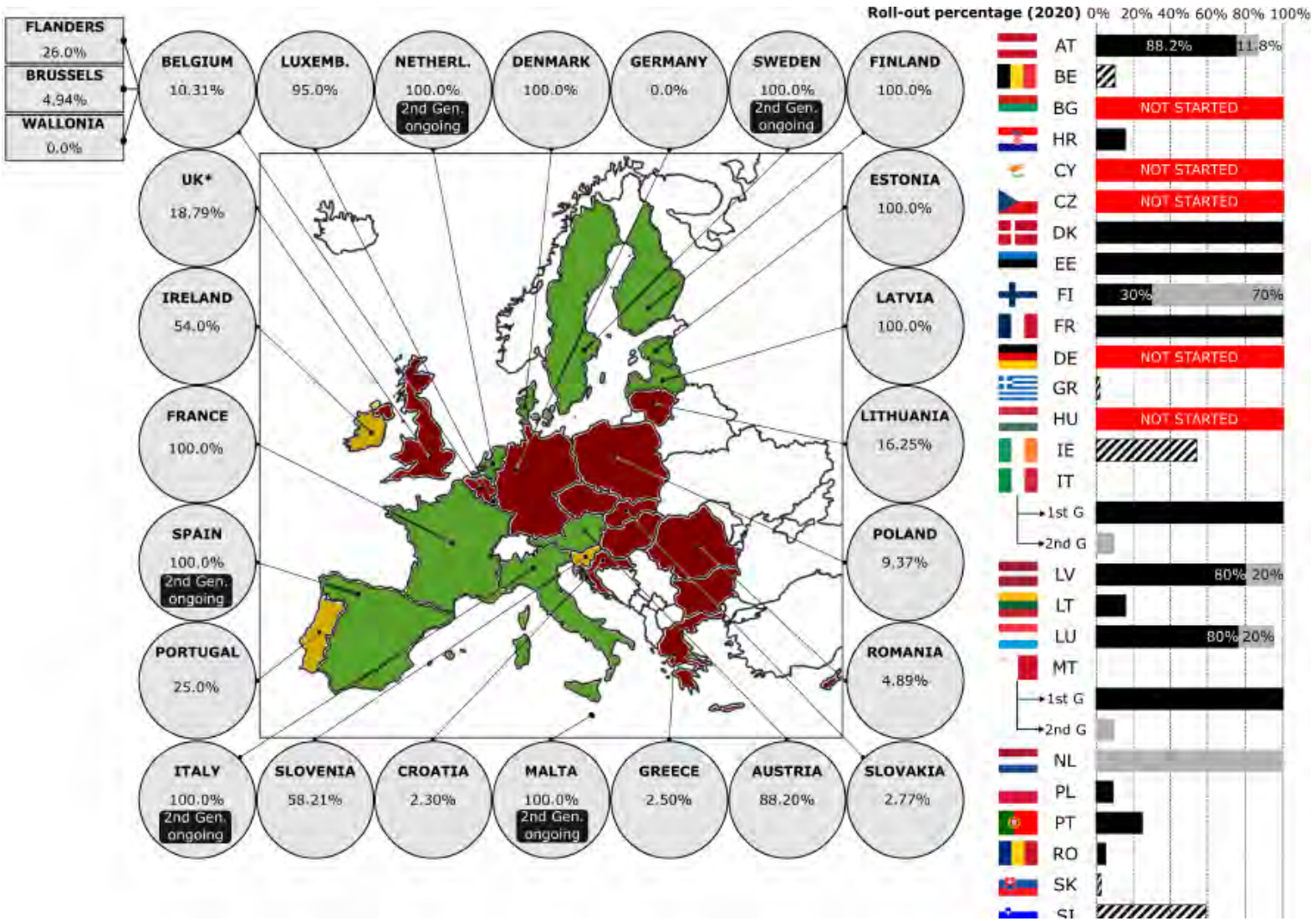
Evolving DSOs roles

- DER growth shifts congestion/voltage issues to distribution grids
- DSOs increasingly procure or activate flexibility for local objectives
- Smart metering/AMI is critical for observing and coordinating millions of endpoints

Traditional roles for DSOs	Emerging roles for DSOs
• Planning (fit-and-forget) • Maintenance/management • Grid outage management • Metering (billing purposes)	• Peak-load management • Congestion management • Reactive power support • Voltage support procurement • Neutral market facilitator • Reduce network investments

Source: https://www.irena.org/-/media/Files/IRENA/Agency/Publication/2019/Feb/IRENA_Landscape_Future_DSOs_2019.pdf

EU disparity: rollout, communication, and readiness are uneven



What to notice:

- Penetration varies from near-0 to ~100%.
- Technology mix differs (PLC vs wireless).
- Data governance & privacy differ by country/DSO → affects access and services.
- Therefore: capability gaps and "disparity" must be quantified, not assumed.

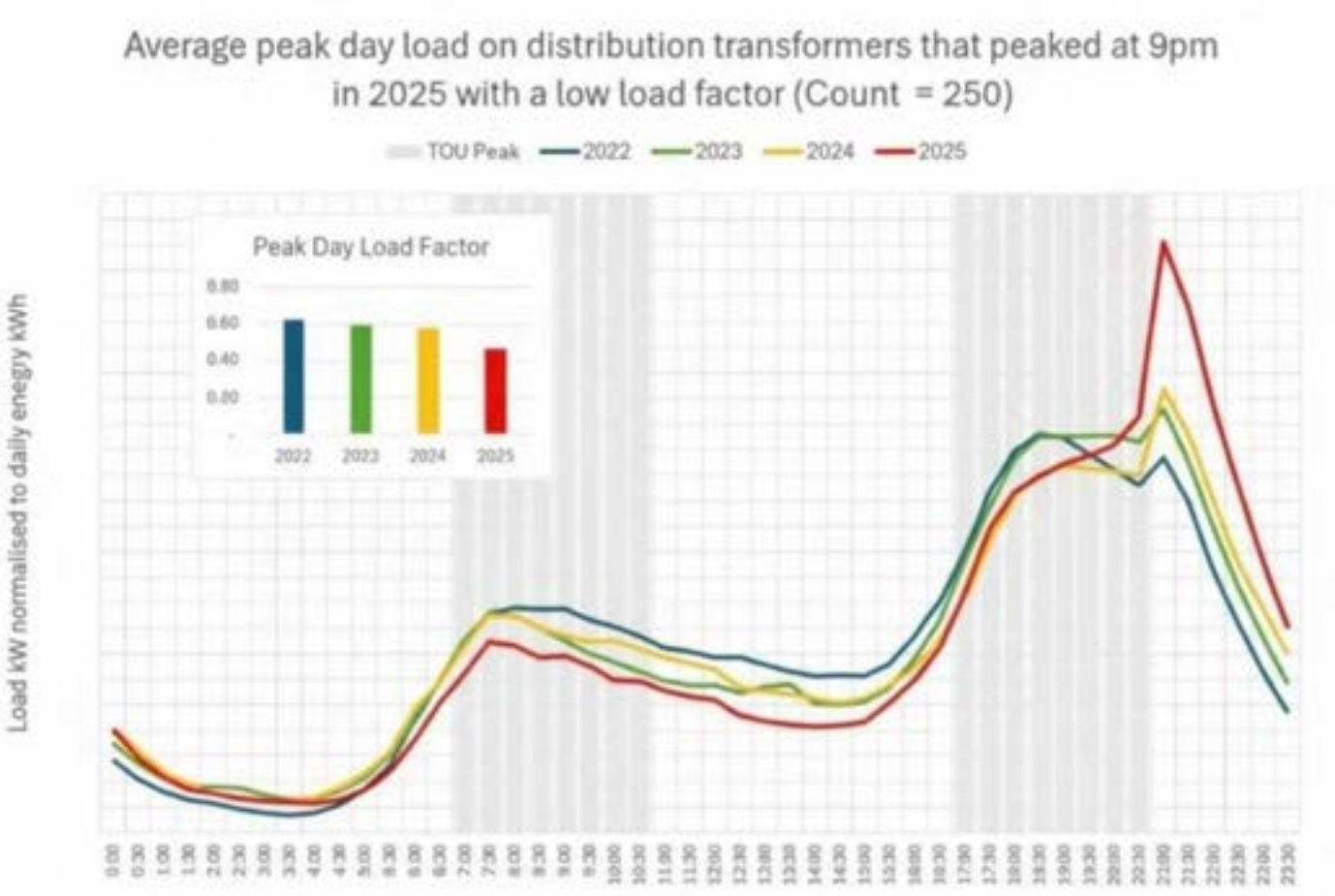
Same "smart meter penetration %" can hide very different capabilities

Kippke Salomon, M. A., Hashmi, M. U., Blavette, A., & Arboleya, P. (2025, June). Smart metering capability towards harnessing distribution network flexibility: gaps and disparity quantification. In *IET Conference Proceedings CP922* (Vol. 2025, No. 14, pp. 2543-2548). Stevenage, UK: The Institution of Engineering and Technology.

Why we need flexibility?

Some additional examples

- **Changing generation mix**: In Ireland the ratio between total non-synchronous generation and the total instantaneous demand exceeds 70% many times over the year
- **High simultaneity** of DERs
- **Consumer rationality**
- **Aging distribution infrastructure**



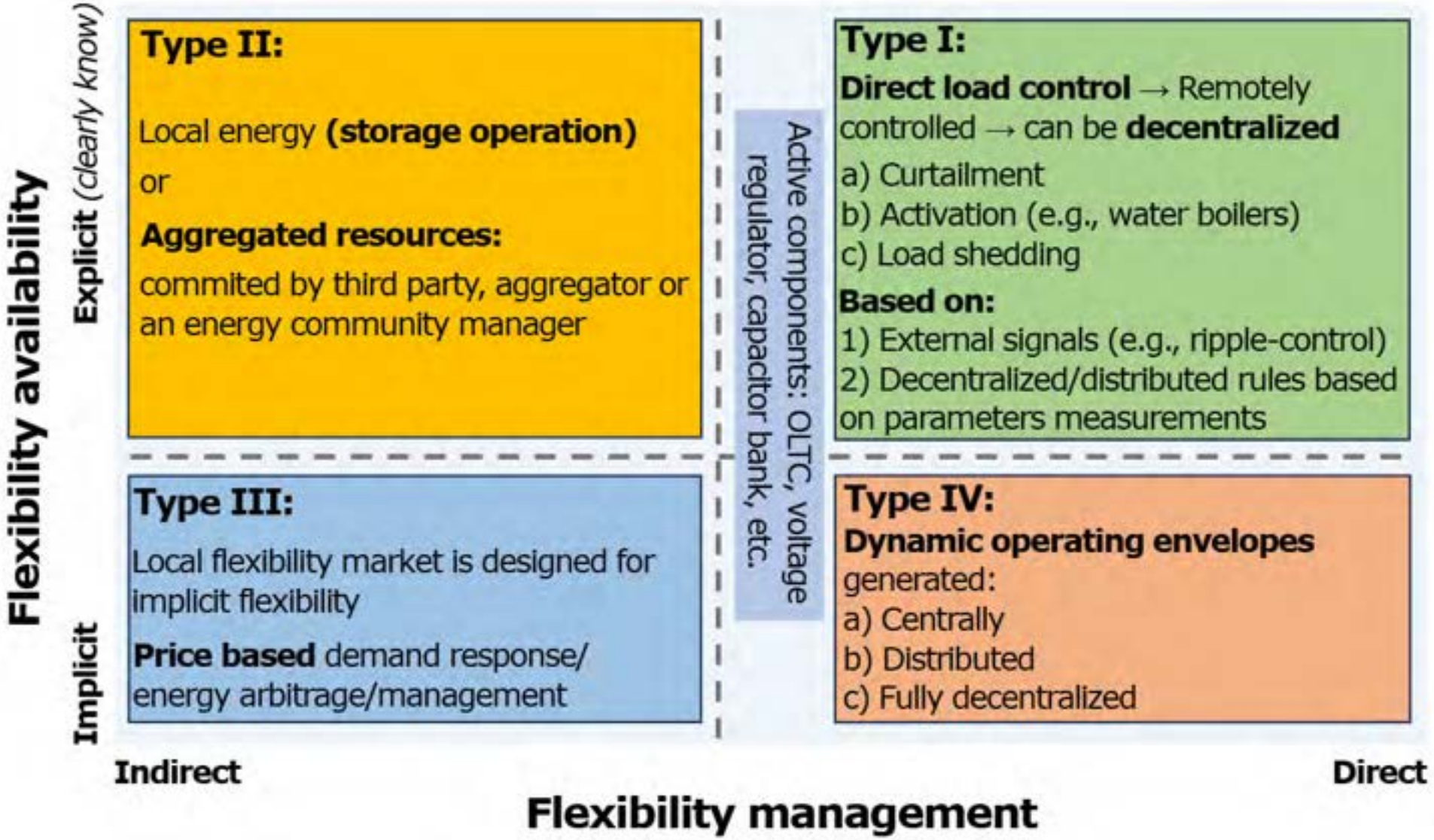
Sources:
[1] https://www.eirgrid.ie/news/renewablesprovided-third-electricity-september
[2] https://www.thepost.co.nz/business/360825632/kiwis-are-shifting-their-electricity-use-not-always-ways-are-helpful

Flexibility taxonomy: availability vs. management

- Availability: do we know the flexibility amount:
  - explicitly (contracted/quantified)
  - implicitly (response uncertain)

- Management: who controls the asset?
  - Direct control: by the system operator
  - Indirect control: dispatch signal generated not by the system operator
    - Aggregator controlled
    - Local control by HEMS/user

Flexibility classification



Type 1 flexibility

stylized representation

Communication/data exchange needs

Type I: High-dependency

→ Activation signal receiving capability in **close to real-time**

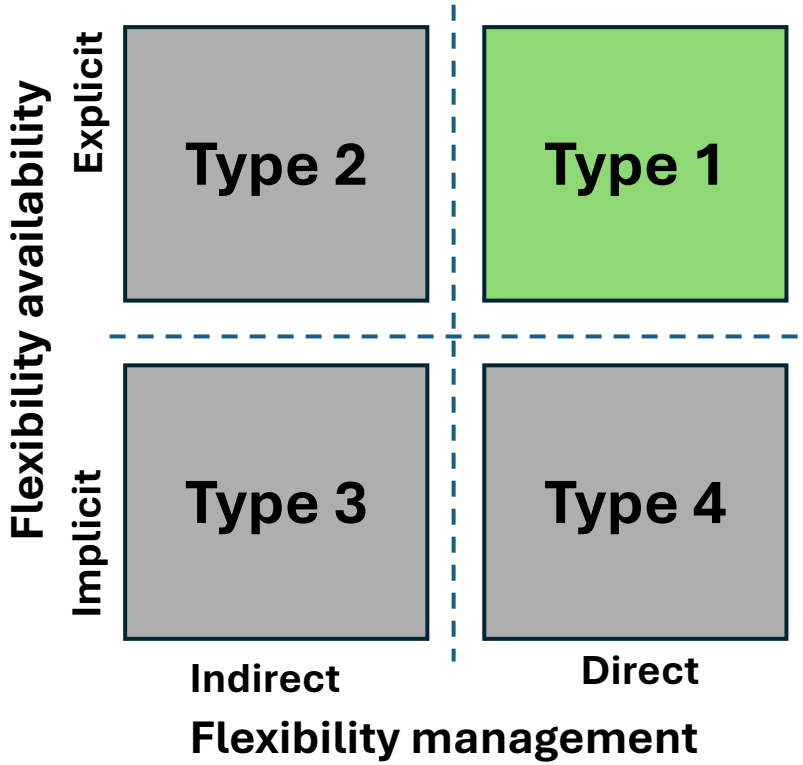
→ Flexibility activation feedback

→ Low latency

Type 1 Flexibility

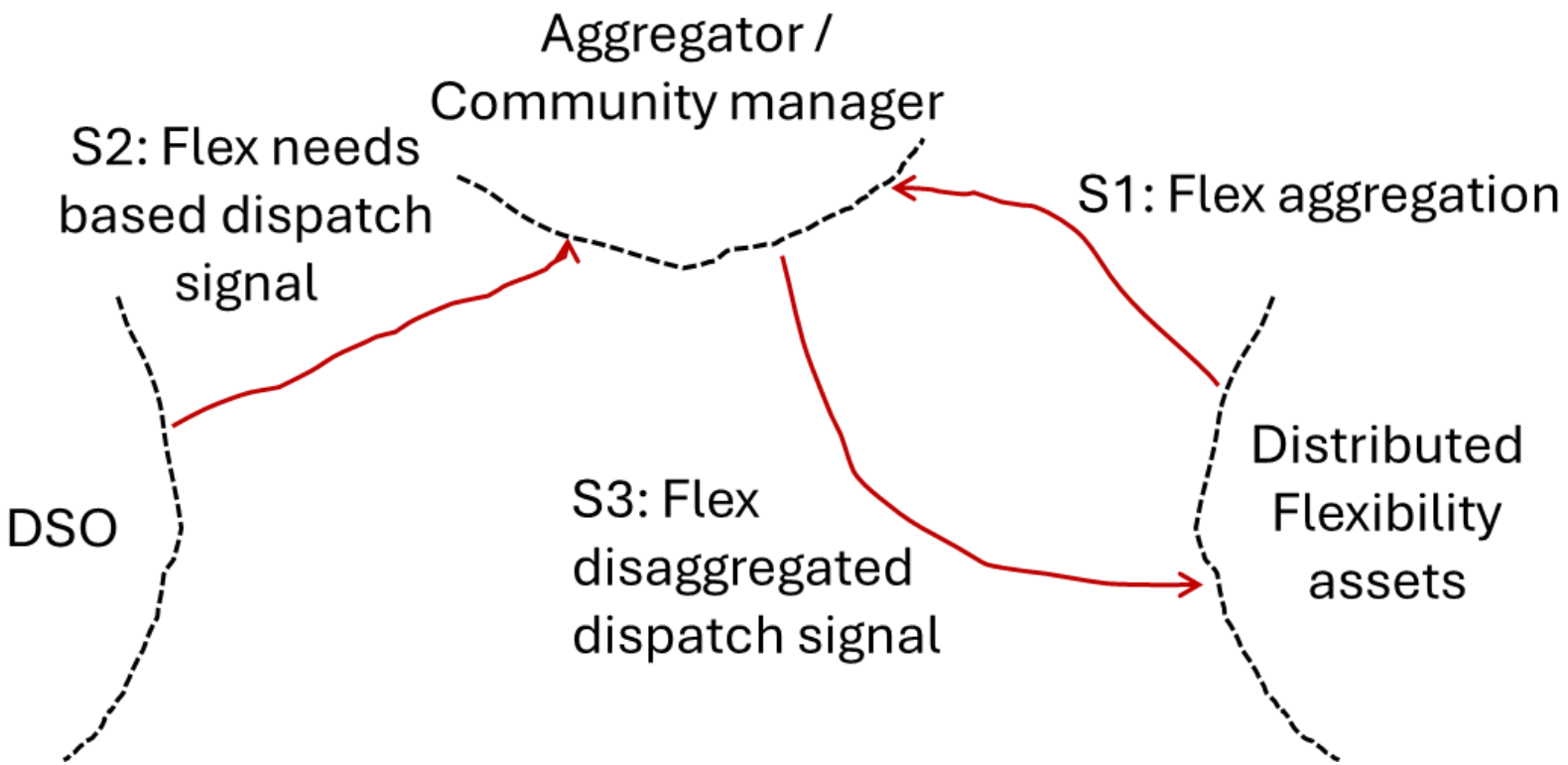
Explicit availability with direct management

- Direct load control
  - DSO operating non-DSO owned assets: Germany's new §14a EnWG regulation (effective 1 January 2024) formally introduces a nationwide framework allowing DSOs to directly limit the power consumption of flexible loads during local grid congestion (for assets > 4.2 kW). Remote control options are:
    - Ripple Control Receivers (FNN-compliant)
    - Modbus TCP
    - OCPP 2.0.1 (cloud-based control)
  - DSO operating directly owned assets → Load tap changers, voltage regulators
- Decentralized/local preset control
  - Based on PCC voltage: Mediterranean DSOs implement Volt-Var local controllers first and introduce Volt-Watt profiles when reactive power alone is insufficient.
  - Spain: Automatic Power Reduction System (SRAP) and Active Demand Response Service (SRAD)



Type 2 flexibility

stylized representation



Communication/data exchange needs

Type II: Mild-dependency

→ Activation signal receiving capability

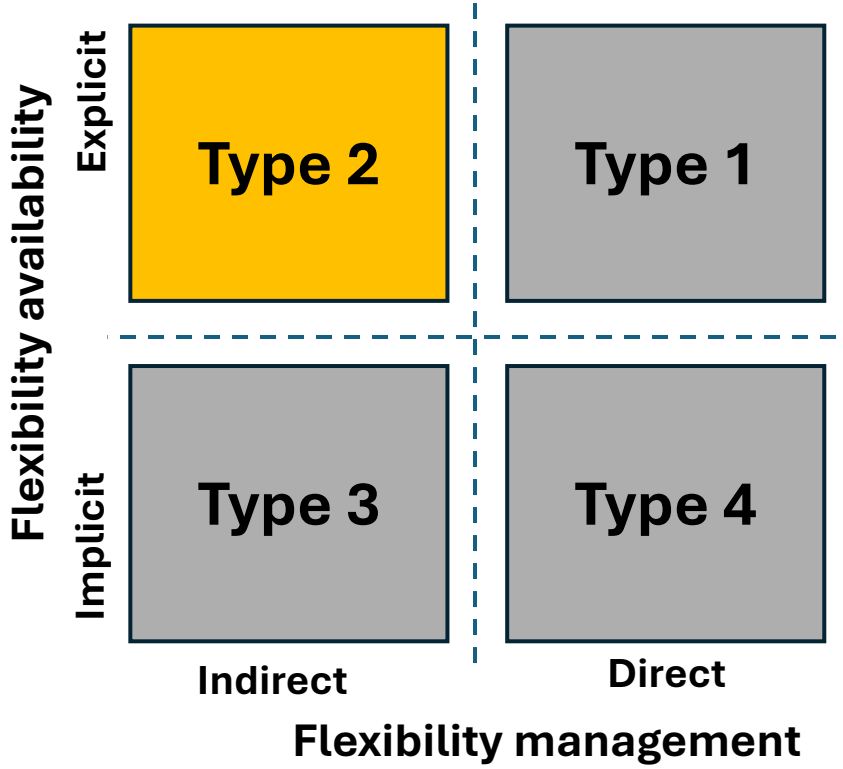
→ Scheduling capacity

→ Performance quantified to be communicated to central hub

Type 2 Flexibility

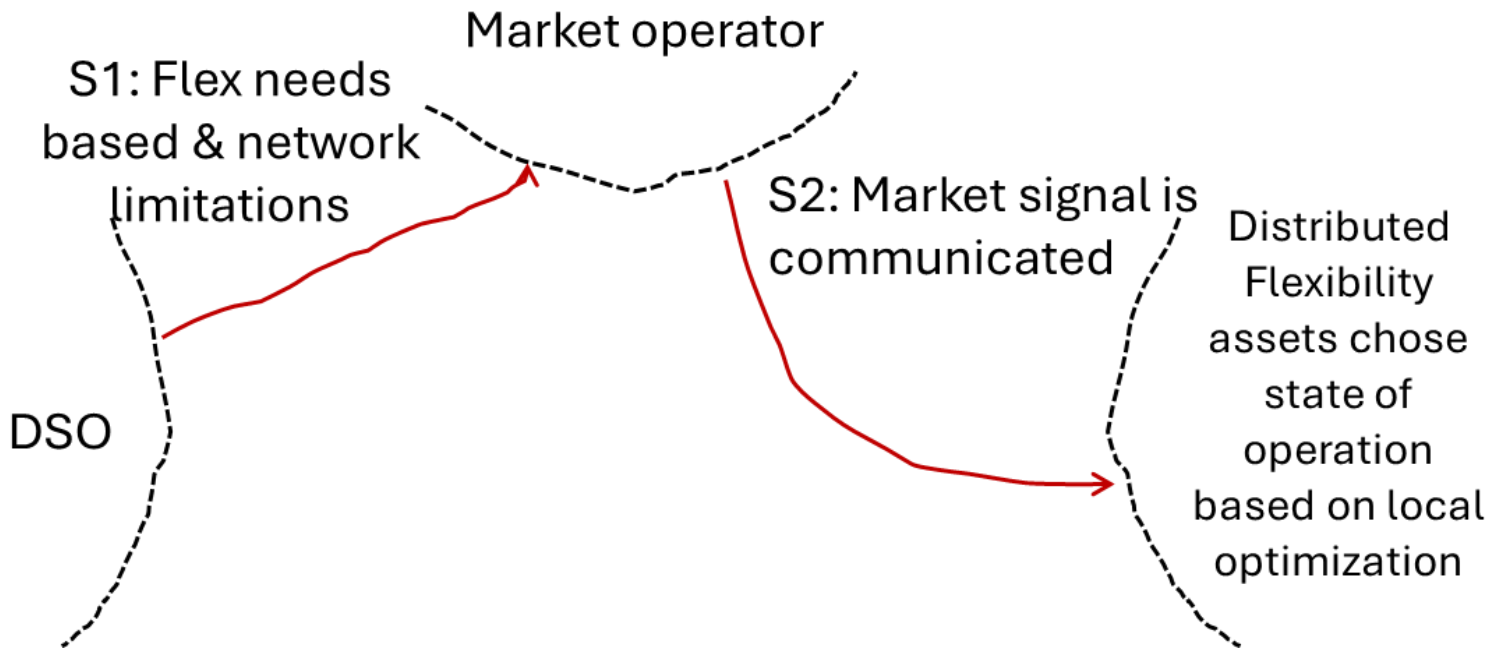
Explicit availability with indirect management

- Sources: Flexibility from storage (batteries, thermal storage) or aggregated resources coordinated by an aggregator/energy community.
- Example of type 2 flexibility:
  - Aggregated resources providing contractually agreed upon grid services
    - Portugal: DSOs procure services from aggregated distributed resources via dedicated flexibility tenders. Aggregators bid their portfolios and commit to deliver contracted activation volumes.
    - Ireland: Irish DSOs use voluntary Flexible Connection Agreements (FCAs) enabling customers and aggregators to provide flexibility on a contracted basis to mitigate local grid congestion. These aggregated resources deliver specific, agreed-upon responses when the DSO triggers them. Services provided:
      - Curtailment on demand
      - Temporary load modulation
      - Reverse power-flow management for renewable hotspots



Type 3 flexibility

stylized representation



Communication/data exchange needs

Type III: Low-dependency

→ Market/price signal receiving

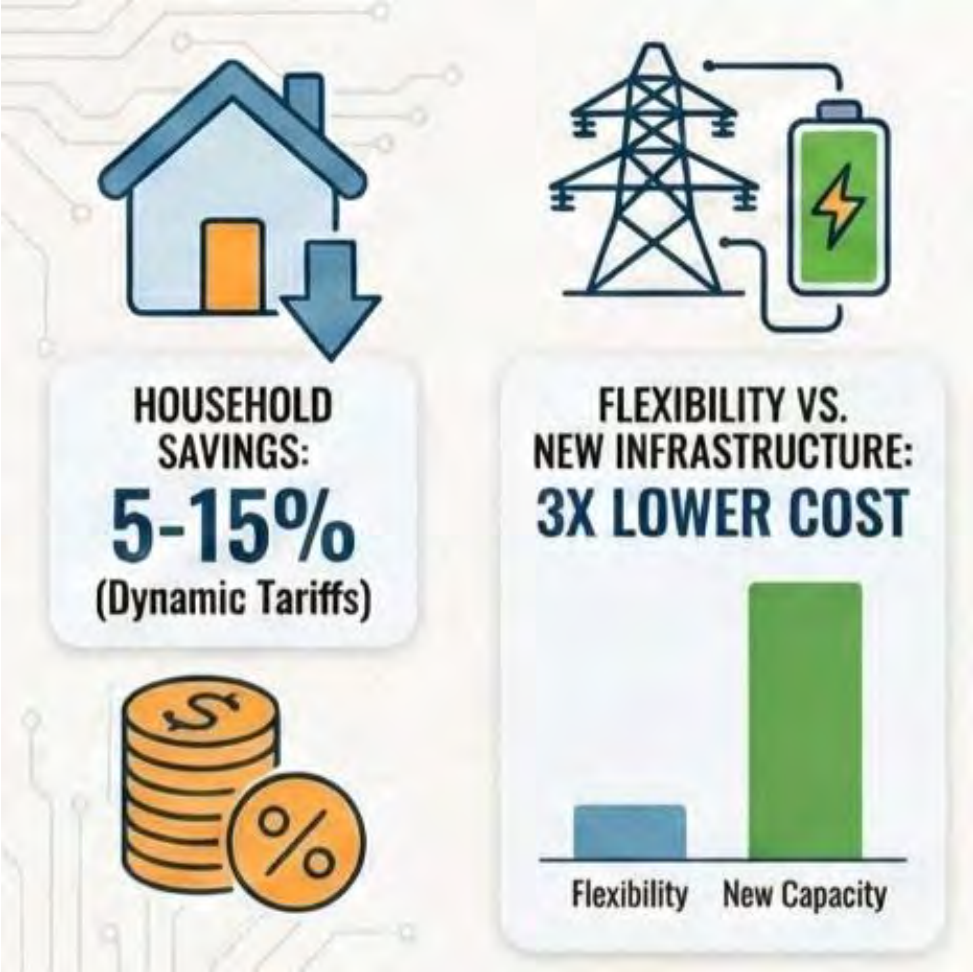
→ Performance quantified to be communicated to central hub

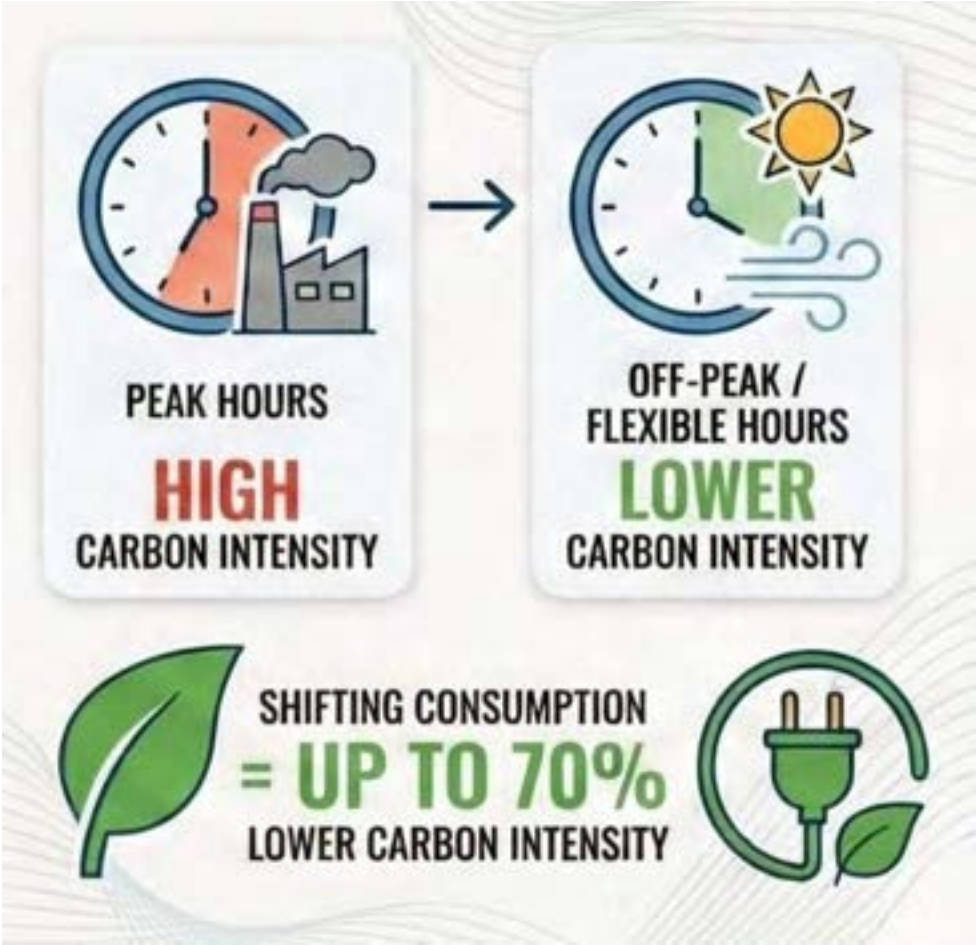
→ Load control decision making via local intelligence (HEMS)

Type 3 Flexibility

Implicit availability with indirect management

- ToU tariffs / Capacity tariff/ critical-peak pricing

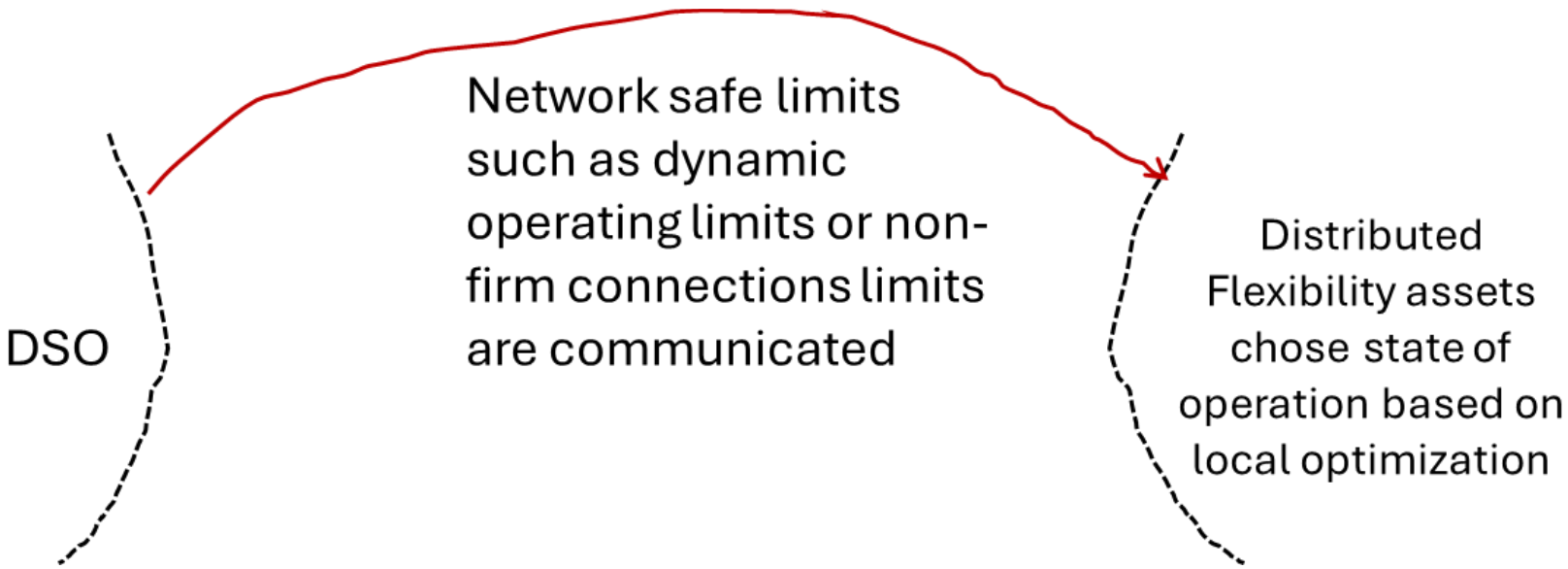




Graphics source: NODES platform

Type 4 flexibility

stylized representation



Communication/data exchange needs

Type IV: Independent

→ Monitoring and control capacity at the connection point

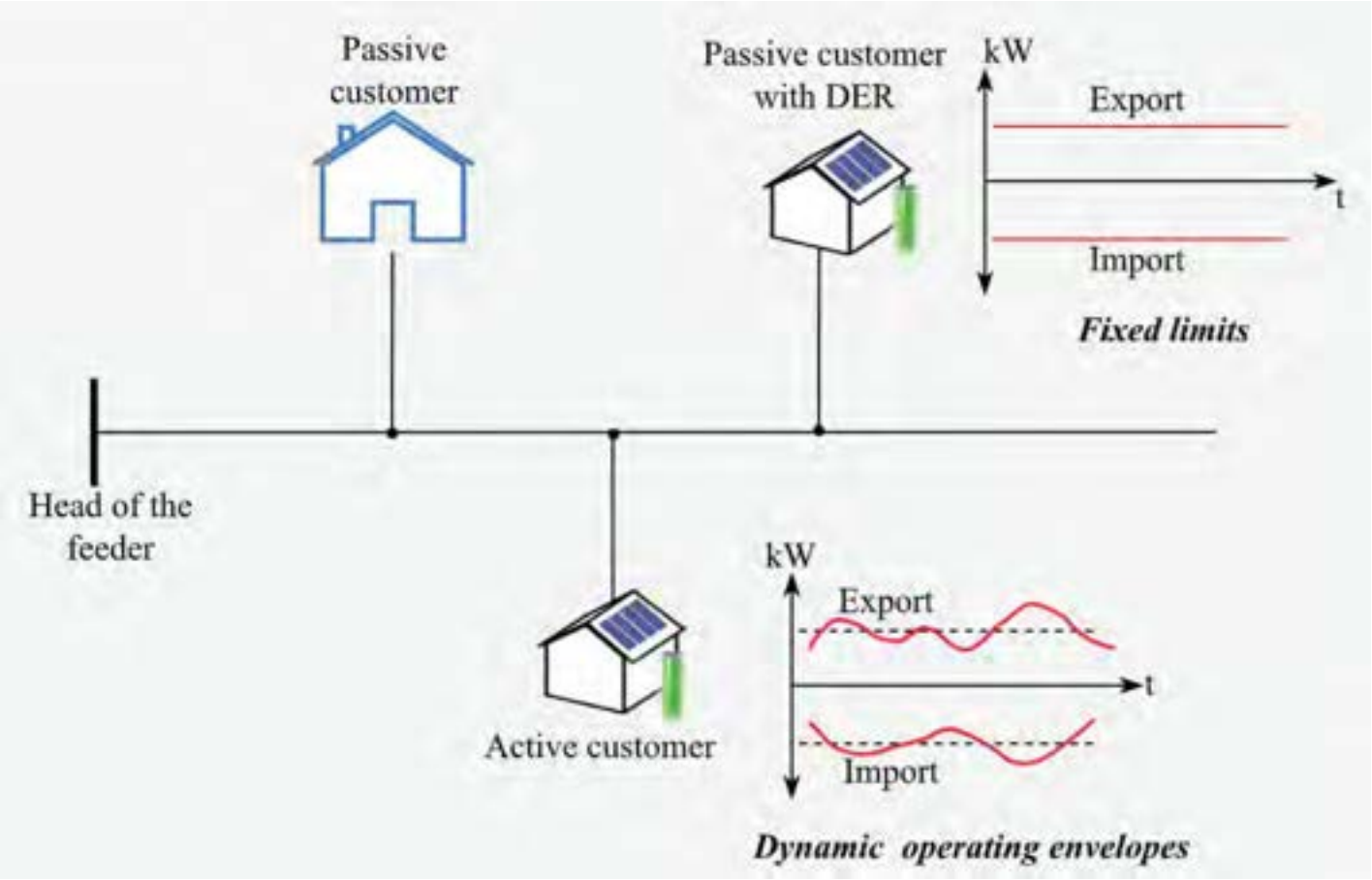
→ Flexibility activation feedback

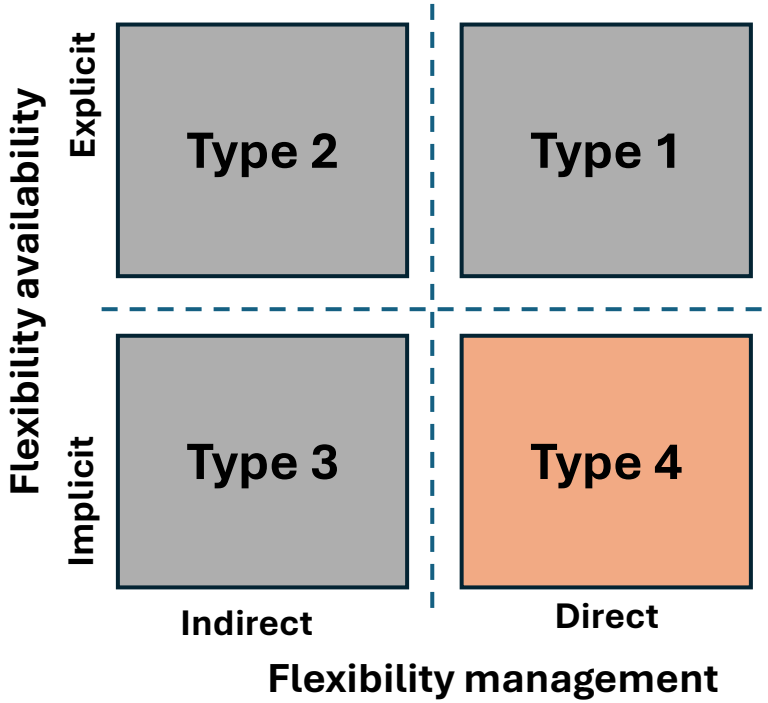
→ Load control decision making via local intelligence (HEMS)

Type 4 Flexibility

Implicit availability with direct management

- Dynamic operating envelopes



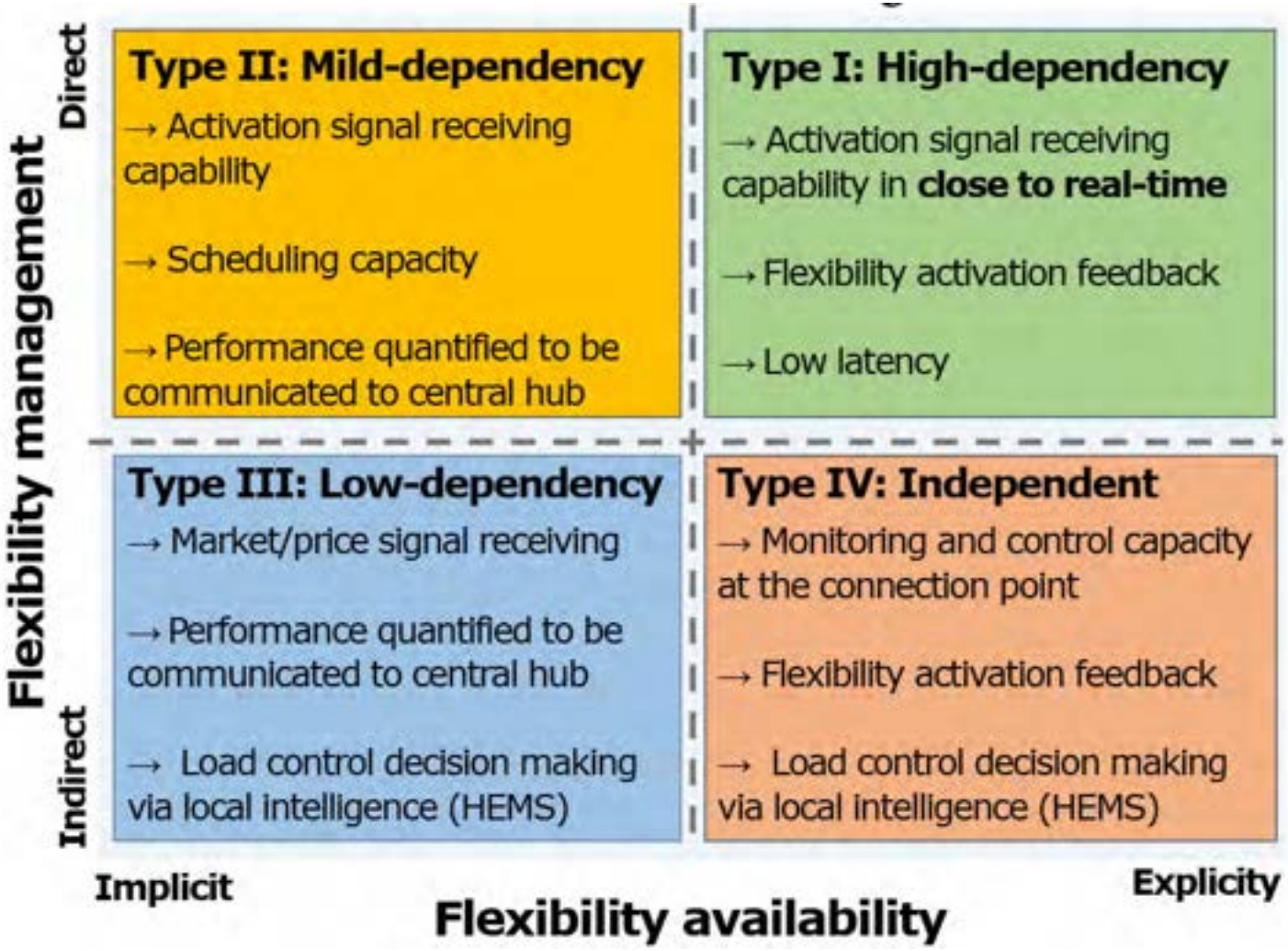


Lankeshwara, Gayan, et al. "A review on network-aware control of distributed energy resources under the dynamic operating envelopes framework." *Renewable and Sustainable Energy Reviews* 217 (2025): 115696.

Summary of flexibility classification

Flex type	Actors	Subtypes	Local optimization possibility	Dedicated hardware
Type 1 – Direct, Explicit	DSO, flex user /aggregator	1A: fully observable real time communication needs 1B: Partially observable 1C: Historical based	No / limited	Not necessarily
Type 2 – Indirect, implicit	DSO, aggregator/ community manager		No / limited	Yes
Type 3 – Indirect, explicit	DSO, market operator, aggregator, flex participant	3A: Real time network state-based market signal 3B: Historical patterns-based market signal generation such as ToU	Yes	Not necessarily
Type 4 –direct, explicit	DSO, flex participant	4A: Real time network state based 4B: Historical data based	Yes	Not necessarily

SM features necessary for harvesting different flexibility types



- Type I needs close-to-real-time signaling, low latency, and activation feedback.
- Type II (DOEs) needs direct control + verification, but not explicit energy commands.
- Type III/IV rely more on signals + performance reporting, often via local intelligence (HEMS).

Communication protocols & latency: why “AMI ≠ real-time control”

First-generation choices shape what flexibility is realistic

What AMI typically optimizes

- Reliable periodic reads for billing
- Cost-effective mass deployment
- Remote operations (firmware, connect/disconnect)

→ Not necessarily deterministic latency

Why latency matters for flexibility

- Type I: seconds-level loop (signal → response → confirmation)
- Type II: envelope updates + compliance monitoring
- Type III/IV: minutes–hours often acceptable

→ Pick the market product to match the infrastructure

Europe’s infra today (simplified):

- Many high-penetration countries use NB-PLC (PRIME / G3-PLC) on CENELEC-A band.
- PLC can suffer from distortion/data loss (worse with high DER), limiting advanced services.
- Second-generation rollouts increasingly consider wireless (GSM/LTE/5G), but scale & cost remain challenges.
- Behind-the-meter DER protocols are fragmented; lack of standardization increases integration cost.

Spain as a concrete example: meter types imply capability tiers

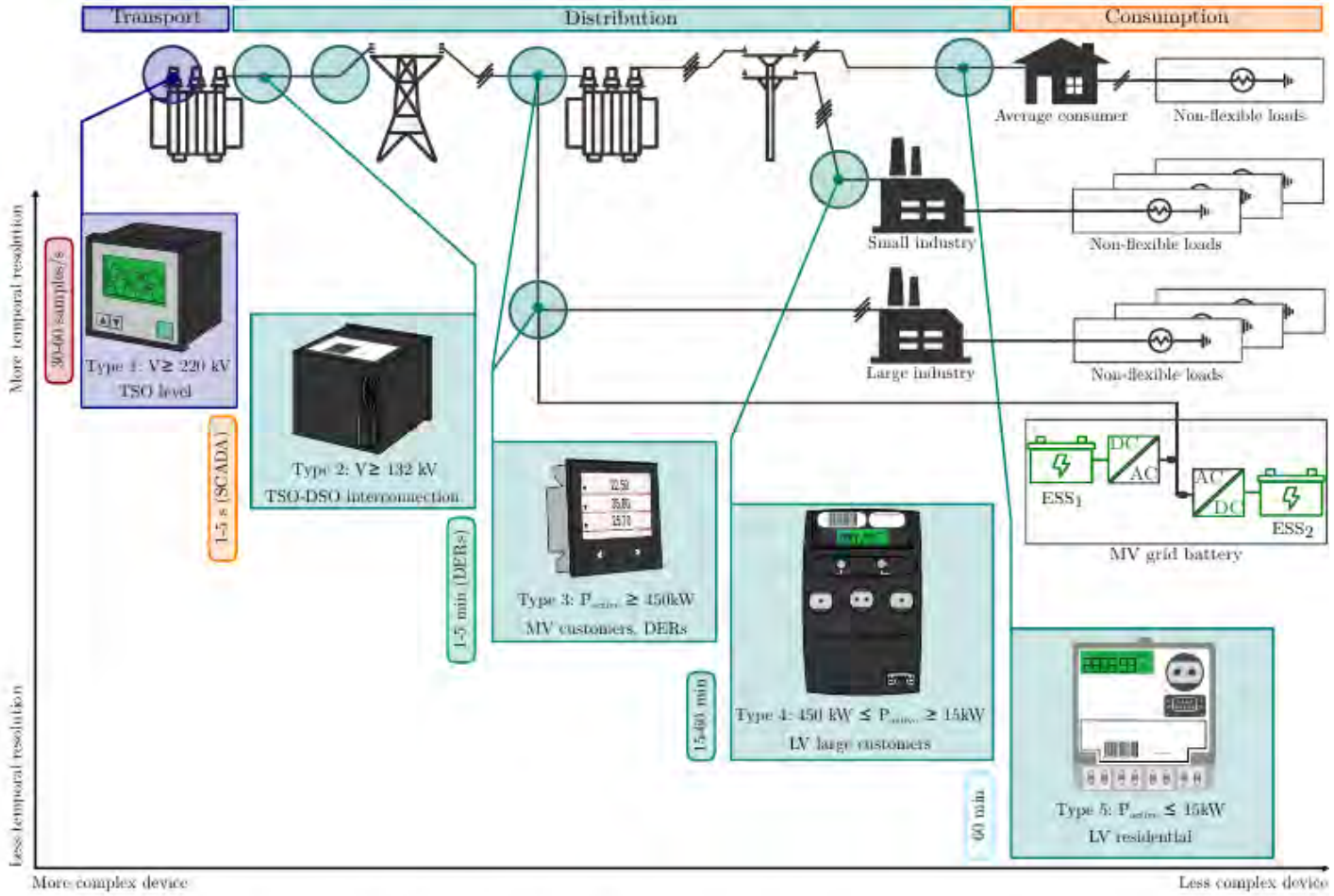


Figure 4.1: Different classification for smart meters according to Spanish Law.

Main points:

- Types 1–2: SCADA-grade metering (transmission / interconnection)
- Type 3: MV industrial + DER points (remote reading required; AMI integration depends on setup)
- Types 4–5: LV endpoints on AMI (designed for mass rollout; PLC/RF common)
- Resolution/latency expectations differ by design.

Policy enablers & market dynamics

Policy creates the “who/what/how”

- Who can provide flexibility (consumers, aggregators, communities)
- What products exist (capacity, energy, DOEs, local services)
- How they are procured (market-based, transparent, non-discriminatory)
- How performance is measured & settled

Market dynamics

A flexibility product is “good” when:

- value is clear (avoided reinforcement, QoS, losses)
- participation costs are low (standard products, automation)
- verification is feasible with existing metering
- incentives align with consumer protection & privacy

A quick map of enablers

- Legal basis for DSOs to procure services (and define products)
- Aggregator participation rules + baseline/settlement
- Data access & interoperability mandates (APIs, consent, neutrality)
- Consumer safeguards (opt-out, transparency)

Policy without capability = paper markets; capability without policy = stranded tech

Handout description

Country / Region	Smart Metering & Communication	Key Stakeholder & Privacy Factors	Type I (Explicit/Direct)	Type II (Implicit/Direct)	Type III (Implicit/Indirect)	Type IV (Explicit/Indirect)
Australia	Mixed penetration by state. Leading in DER adoption (rooftop solar).	AEMO (market operator) and DSOs are highly proactive in managing DER impacts.	Feasible (Proven). Emergency Solar Management in Western Australia is a direct control mechanism to curtail solar remotely.	Feasible (Proven). A global leader in developing and trialing Dynamic Operating Envelopes (DOEs) to manage DER exports.	Feasible. Multiple VPPs are active, aggregating DERs to participate in wholesale energy and ancillary service markets.	Highly Feasible. Advanced inverter standards (AS 4777.2) mandate autonomous voltage response modes.
USA: California	High penetration of smart meters. Advanced communication networks in place.	Investor-Owned Utility (IOU) model. Strong state regulator (CPUC). FERC Order 2222 is a key federal driver.	Feasible. Long history of direct load control programs for air conditioners (e.g., Critical Peak Pricing with enabling tech).	Emerging. IOUs are beginning to explore and pilot DOEs as an alternative to curtailment and costly upgrades.	Feasible (Proven). FERC Order 2222 mandates that TSOs/ISOs allow DER aggregations to participate in wholesale markets.	Highly Feasible. California's Rule 21 mandates that all new DERs must be "smart" inverters with autonomous grid-support functions.
China	Very High penetration from massive, state-led rollouts. Mix of PLC and advancing wireless/5G capabilities.	Dominated by state-owned utilities (e.g., State Grid). Top-down control. Privacy concerns are secondary to grid operational goals.	Highly Feasible. The centralized, state-controlled utility model is perfectly suited for large-scale, direct load control and curtailment.	Emerging. Technologically possible, but the command-and-control model may favor explicit instructions over envelopes.	Challenging/Emerging. Markets are nascent and heavily regulated. The primary model is direct utility management, not market-based procurement.	Feasible. As a major manufacturer of DERs, China can easily mandate technical standards for autonomous grid support functions.

What do we learn and what next?

Framework + roadmap + transfer learning

Contribution A: operational flexibility classification framework

A decision-oriented taxonomy that links:

- Flexibility types (I–IV)
- Implementable solutions (AMI, HEMS, aggregators, DOEs)
- Enablers: technology + policy + regulation

→ Enables “what is feasible here and now?” comparisons

Contribution B: next-step roadmap + transfer learning

Given a country/DSO starting point (penetration, comms, data rules), recommend:

- the most obvious next flexibility step
- prerequisite upgrades (telemetry, APIs, settlement)
- what can be reused from forerunners (products, interfaces, governance)

A simple “next step” ladder

1. Visibility
Get reliable profiles + device inventory

2. Indirect incentives
Type III/IV pilots (low latency not required)

3. Verification
Baselines + settlement + auditing at scale

4. Operational envelopes
Type IV DOEs + compliance monitoring

5. Near-real-time control
Type I where justified (low-latency comms)

Summary

- Flexibility classification framework based on
  - Availability : (known or not known)
  - Management : (by DSO or not by DSO)

- Examples of flexibility frameworks

Day 1: Assignment

Day Assignment 1 – flexibility value chain stakeholders in your allocated country with examples of national stakeholders: DSO, TSO, aggregator, market operator, flexibility provider, consumers, prosumers, bulk generation companies.

Classify and discuss the flexibility types (I, II, III, IV) implementable in your allocated country.

Any questions?

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6) Metering requirements by customer class (Spain): TM vs TG

Telemedida (remote reading) ≠ Telegestión (remote management)

Table 4.1: Meter classification by types according to Spanish regulation. TM: *telemedida* (remote reading). TG: *telegestión* (remote management).
Legend: ✓ mandatory. ● optional. ✗ not applicable.

TM/TG flags

	Scope	Use case	Characteristics	TM	TG
1	≥ 220 kV (HV)	Connection points for large generators and HV consumers	- CT/VT connected - Class 0.2S - SCADA integrated	✓	✗
2	≥ 132 kV (MV/HV)	TSO/DSOs interconnections	- Type 1 similar - Interconnection metering - SCADA integrated	✓	✗
3	≥ 450 kW (MV)	DER plants, industrial consumers with dedicated MV	- CT-connected meters - Class 1 or 0.5S - SCADA integrated	✓	●
4	1-450 kW (LV)	Commercial or residential buildings, high-consumption households	- Direct three-phase - Up to 100 A - PLC/RF comm	✓	✓
5	≤ 15 kW (LV)	Standard residential	- Single or three-phase - MID Class B - PLC/RF comm	✓	✓

In short:

TM enables data acquisition at distance; TG adds remote operations (configuration, events, connect/disconnect), which is necessary for scaling flexibility programs across LV endpoints.

7) Putting it together: which flexibility types are feasible where?

Meter capability tiers × flexibility quadrants

Interpretation matrix (conceptual):

Meter type	Type I	Type II	Type III	Type IV
Type 1–2 (HV/TSO link)	✓✓	✓	—	—
Type 3 (MV DER/industrial)	✓	✓	○	○
Type 4 (LV large)	—	○	✓	✓
Type 5 (LV residential)	—	○	✓	✓
Behind-the-meter (HEMS/DER gateway)	○	✓	✓	✓

✓✓ strong fit ✓ feasible ○ conditional — typically not

Use this matrix to design “products that fit the infrastructure”